ARIA

DISTRIBUTED EDGE AI ARCHITECTURE

LAYER 1

Companion Device

ESP32

Push Button (GPIO 0)
Start/stop conversation

ST7789 TFT Display
240x320 SPI, word-wrapped responses

4x RGB PWM LEDs
Sine-wave breathing animations

WiFi State Machine
Auto-connects to Pi hotspot

WebSocket Client
Bidirectional state sync + keepalive

WIFI WEBSOCKET

← state →

← text →

/WS/PENDANT
/WS/AUDIO

LAYER 2

Edge AI Brain

Raspberry Pi 5

FastAPI Server
3 WebSocket channels, REST API, session management

GPT-5.2 Agentic Brain
10 tools, 6-iteration reasoning loop, 256 token responses

OpenAI Realtime API
24kHz voice-to-voice, server VAD, persistent sessions

Whisper STT
Ambient transcription + hallucination filtering

3-Tier TTS
ElevenLabs voice clone > OpenAI TTS > espeak

Memory Engine
Auto-extracts facts every 15

WIFI WEBSOCKET

← events →

← chat →

/WS/DASHBOAR

LAYER 3

Live Dashboard

Browser

Animated Waveform
State-driven color, amplitude, speed

Live Transcript Feed
User, ARIA, Ambient speakers

Conversation History
Full chat log with timestamps

Tool Activity Log
Real-time tool calls + results

Memory Viewer
What ARIA has learned about you

min, top 25 in context

SQLite (5 tables)
Transcripts, memories, conversations, reminders, calendar

3D-Printed Enclosure
Custom Autodesk Inventor design

Bluetooth Audio
AirPods I/O via ALSA (arecord/aplay)

Voice + Text Input
Web Speech API, zero polling

Button Press → **Voice Stream** (24kHz) → **GPT-5.2** reasons + tools → **ElevenLabs** voice clone → **AirPods** output

7 APIS INTEGRATED

- OpenAI Realtime
- GPT-5.2
- Whisper
- OpenAI TTS
- ElevenLabs Flash v2.5
- Brave Search
- X / Twitter API

3	10	7	3	5	$30
COMPUTE LAYERS	AGENTIC TOOLS	API INTEGRATIONS	WEBSOCKET CHANNELS	DATABASE TABLES	TOTAL HARDWARE COST